

Mosyle MDM Lab

Configuration Profile Deployment – Personal Lab Documentation

Overview

This lab documents hands-on experience configuring and deploying Mobile Device Management (MDM) profiles using Mosyle Manager, an Apple-focused MDM platform widely used in K-12 education environments. The goal was to build practical familiarity with profile-based device management — the same workflow used in real school and enterprise Apple deployments — covering device tagging, network authentication, application restrictions, and browser policy enforcement.

Platform	Mosyle Manager (personal trial account)
Devices	macOS (enrolled via .mobileconfig files)
Scope	Multiple simulated locations with assignment-based targeting

1. MDM Enrollment

Enrollment began by downloading two signed **.mobileconfig** profiles to the macOS device. These were installed via System Settings > Device Management, surfacing a prompt that clearly listed all rights granted to the MDM server — including remote wipe, configuration profile management, and application/media management. Reviewing these rights before accepting is a critical security step and mirrors the trust decision administrators must communicate to end users.

Key detail: Profiles were signed by *.mosyle.com, confirming authenticity prior to enrollment.

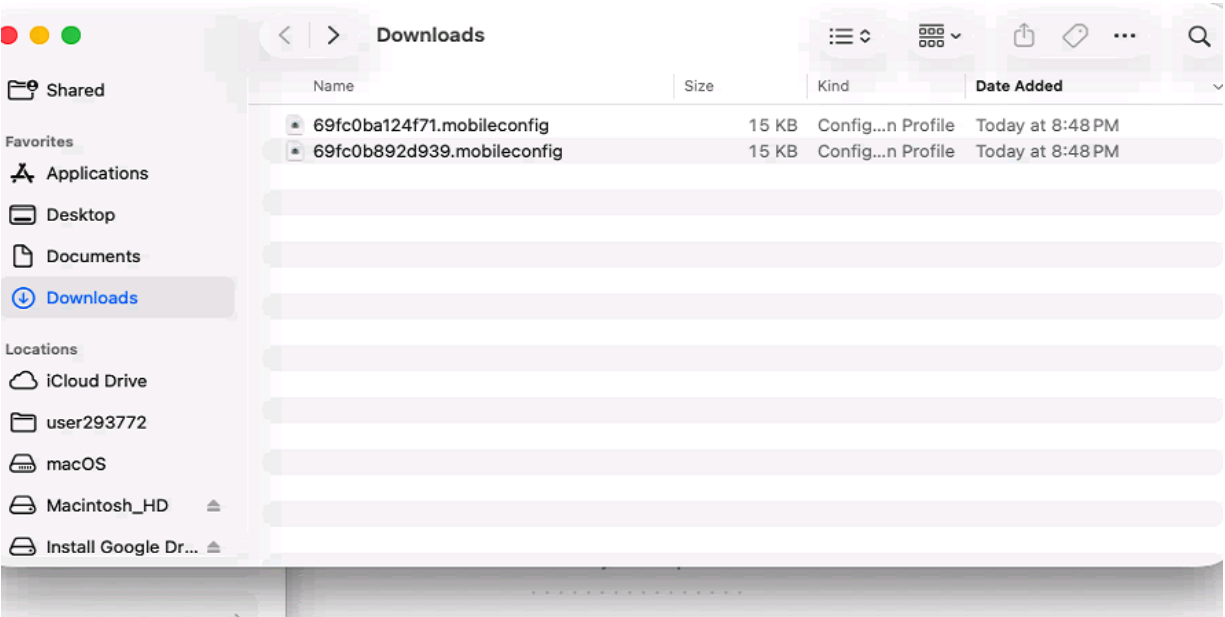


Fig. 1 – Downloaded .mobileconfig enrollment profiles in Finder

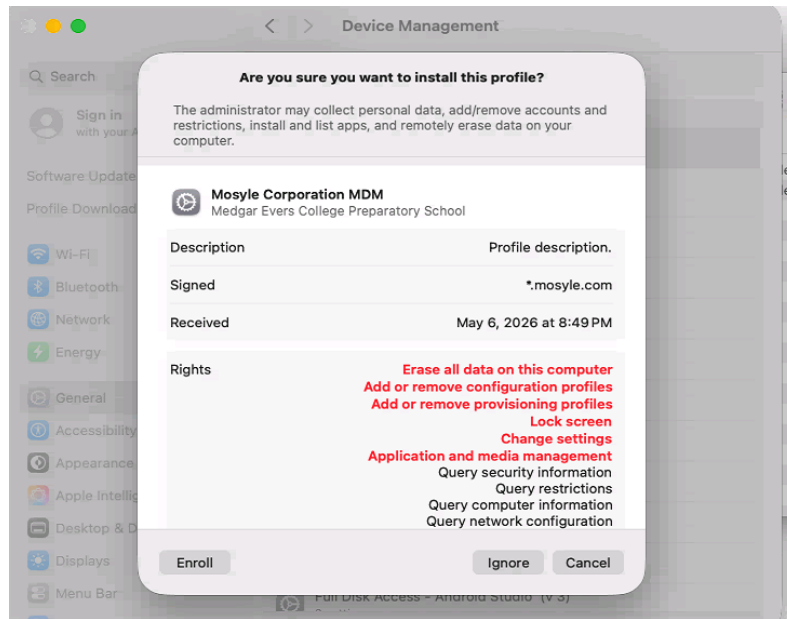


Fig. 2 – MDM enrollment prompt detailing rights granted to the Mosyle MDM server

2. Digital Device Tag Profile

Profile Name: DeviceTagProfile

A Digital Device Tag profile was configured to use the device's **serial number** as its identifier within the Mosyle console. This enables scalable asset tracking and ensures every policy push, alert, and compliance event is tied to a specific physical device — no manual inventory entry required.

- Tag value source: Device Serial Number
- Admin Direct Link Reader: Enabled
- Assignment: Applied to 2 specific locations

Digital Device Tag

Profile Name *

DeviceTagProfile

Select which information to use as the Digital Tag value *

Device Serial number

☒ Enable Admin Direct Link Reader

Profile Assignment
Select the users and devices to receive this profile.

Select Location Type

Apply to specific Locations

Select Location(s)
When setting up your management profile, select one or more Locations to be assigned to registered users at the Location(s) chosen.

Select locations
2 location(s)

☐ Allow Location Leaders to view this profile

All current and future Devices Limbo(remove)

This profile will be assigned to all devices limbo on Device Enrollment.

+ Add Assignment

Fig. 3 – Digital Device Tag profile using serial number as the tag value

3. Wi-Fi Authentication Profile

Profile Name: WifiAuthProfile

A Wi-Fi configuration profile was created to push network credentials to managed devices automatically, eliminating manual configuration by end users. MAC address randomization was explicitly disabled — in managed fleets, consistent MAC addresses are required for network-layer visibility, DHCP reservations, and 802.1X enforcement.

Setting	Value
SSID	Senya
Security Type	WPA/WPA2 Personal
Auto Join	Enabled
Hidden Network	Disabled
Legacy Hotspot	Disabled
Disable Captive Network Detection	Enabled (iOS)
Disable MAC Address Randomization	Enabled

Network information

Profile Name *

WifiAuthProfile

SSID (network name) *

Senya

Security Type

WPA/WPA2 Personal

Password

[Show password](#)

☐ Hidden Network
Enable if target network is not open or broadcasting

☒ Auto Join
When checked the network is auto-joined. If not the user has to tap the network name to join it.

☐ Legacy Hotspot
Enable if target network is treated as hotspot

☒ Disable Captive Network Detection (iOS only)
Do not show Captive Network Assistant

☒ Disable MAC Address Randomization ⓘ

Proxy HTTP

None

Fig. 4 – Wi-Fi Authentication profile with Auto Join and MAC randomization disabled

4. Device Restriction Profile

Profile Name: DeviceRestrictions

A restriction profile was applied targeting specific built-in app categories and system features. Mosyle's restriction builder presents an icon-based selector covering native Apple apps and system functions, mirroring how schools lock down student devices while leaving productivity tools accessible.

- Application time enforcement: Fulltime (24x7)
- Assignment: Applied to 2 specific locations
- Restrictions span system settings, media apps, and communication features

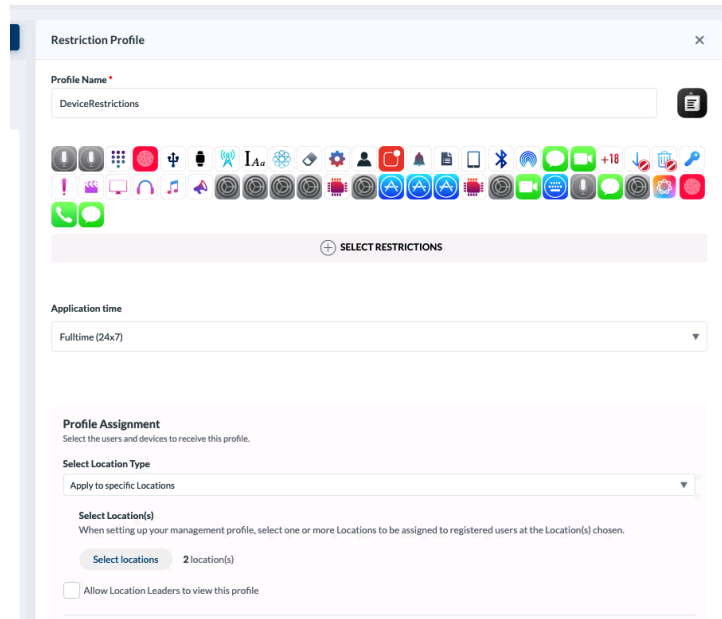


Fig. 5 – Device Restriction profile with app-level restrictions configured

5. Safari Management Profile

Profile Name: SafariManagementProfile

A Safari Management profile was configured across two tabs — Settings and Restrictions — to enforce homepage policy and browser behavior on all managed devices.

Settings Tab

The new tab start page was set to **Home** with the homepage URL pointing to <https://keenantech.xyz>. Assignment scope was set to all current and future locations, meaning any newly added location inherits this policy automatically.

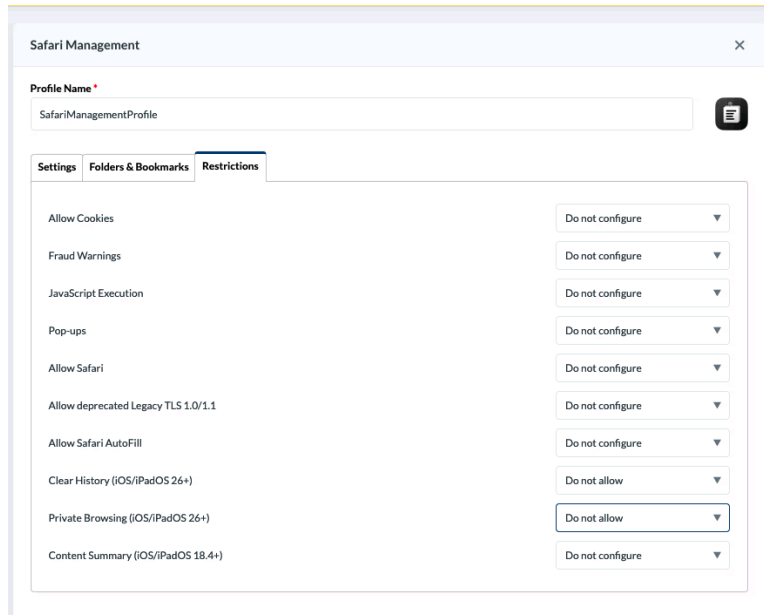


Fig. 6 – Safari Settings tab: homepage enforced to keenantech.xyz across all locations

Restrictions Tab

Two restrictions were actively enforced. All other settings were left at *Do not configure* to avoid overriding system defaults unnecessarily.

Restriction	Setting	Rationale
Clear History (iOS/iPadOS 26+)	Do not allow	Preserves audit trail for content filtering
Private Browsing (iOS/iPadOS 26+)	Do not allow	Prevents bypass of web content monitoring
All other settings	Do not configure	Defers to system default — no unnecessary overrides

Safari Management

Profile Name *
SafariManagementProfile

Settings Folders & Bookmarks Restrictions

New tab Start Page (iOS/iPadOS 26+)
Home
Homepage URL https://keenantech.xyz

Profile Assignment
Select the users and devices to receive this profile.

Select Location Type
Apply to all current and future Locations
☐ Allow Location Leaders to view this profile

No users or devices assigned.

+ Add Assignment

☐ Do not auto-install the profile after saving
☐ Do not apply to devices in Shared Device Groups when a user logs in

Fig. 7 – Safari Restrictions tab: Private Browsing and Clear History set to Do not allow

Key Takeaways

- MDM profiles enable scalable Apple device management — one profile configures hundreds of devices without physical access.
- Assignment scoping (specific locations vs. all locations) prevents over-applying policies to unintended devices.
- The distinction between "Do not configure" and "Do not allow" is meaningful: the former preserves defaults, the latter actively enforces a restriction users cannot override.
- Disabling MAC address randomization on managed devices maintains network-layer visibility needed for 802.1X and DHCP policy.
- Mosyle and Jamf share the same underlying Apple MDM protocol — experience in one transfers directly to the other.

Tools: Mosyle Manager | Apple MDM Protocol | macOS System Settings | .mobileconfig profiles